Fengshuo Liu

Education

2021-present	Ph.D. candidate (Cancer and Cell Biology)	Baylor College of Medicine, Houston, TX
2019-2021	M.S. (Immunology)	New York University, New York, NY
2014-2018	B.S. (Biological Science)	Xi'an Jiaotong-Liverpool University, China

Honors and Awards

2024	Claude W. Smith Outstanding Research Performance (One awardee per graduate program annually), Baylor College of Medicine
2017-2018	Outstanding Student of Xi'an Jiaotong-Liverpool University
2017	National Scholarship, China
2016-2018	National Student Grant, financial support for student with outstanding academic performance
2016-2017	Gold Prize , International Genetically Engineered Machine (iGEM)
2015-2016	Silver Prize , International Genetically Engineered Machine (iGEM)

Research Experiences

2021-present Graduate Research Assistant at Dr. **Xiang H.-F. Zhang**'s Lab
Cancer and Cell Biology Program in Graduate School of Biomedical Science
Baylor College of Medicine, Houston, TX

- Using high-dimension analysis, profiles the cancer and immune system's dynamic changes in response to tumor progression.

2019-2021 Graduate Research Assistant at Dr. **Reizis Boris**'s Lab
School of Medicine
New York University, New York City, NY

- Plasmacytoid dendritic cells kinetics in response to viral infections and tumor progressions through mouse model.

2017-2019 Undergraduate Research Assistant at Dr. **Sung-Kay Chiu**'s Lab
Department of Biological Science
Xi'an Jiaotong-Liverpool University, Suzhou, China

- Applied recombined protein into the mammalian cell medium and evaluated its effects on cell proliferation and migration.

2015-2017 iGEM member at Xi'an Jiaotong-Liverpool University
International Genetically Engineered Machine (iGEM) .
Synthetic biology team, Department of Biological Science
Xi'an Jiaotong-Liverpool University, Suzhou, China

- Developed the random mutation cassette by random mismatch mechanism of the Qß viral genome.

First-authored publications

1. Rivas, C. H.*, **Liu, F.***, and Zhang, X.H.-F.* (2024) *The Roles of Myeloid Cells in Breast Cancer Progression*. In T. Sørlie, & RB. Clarke (Eds.), *A Guide to Breast Cancer* (Chapter 19). ***Springer Cham*** https://doi.org/10.1007/978-3-031-70875-6_19 (*equal contribution)
2. **Liu, F.*,** Zhang, X.H.-F.*, & Yang, Y.* (2024) *Simulation of CRISPR-Cas9 Editing on Evolving Barcode and Accuracy of Lineage Tracing.* ***Scientific Report*** 14, 19213. https://doi.org/10.1038/s41598-024-70154-7 (*equal contribution)
3. Li, Y.*, **Liu, F**.*, Cai, Q.*, Deng, L.*, Ouyang, Q., Zhang, X. H., Zheng, J. (2025) *Invasion and metastasis*

in cancer: molecular insights and therapeutic targets. ***Signal Transduction and Targeted Therapy*** 10(1): 57 https://doi.org/10.1038/s41392-025-02148-4 (*equal contribution)

4. **Liu, F.***, Ding, Y.*, Xu, Z.*, Hao, X., Pan, T., Miles, G., Wu, Y.-H., Liu, J., Bado, I. L., Zhang, W., Wu, L., Gao, Y., Yu, L., Edwards, D. G., Chan, H. L., Aguirre, S., Dieffenbach, M. W., Chen, E., Shen, Y., Hoffman, D., Becerra-Dominguez, L., Rivas, C. H., Chen, X., Wang, H., Gugala, Z., Satcher, R. L., Zhang, X. H.-F. (2025) *Single-cell profiling of bone metastasis ecosystems from multiple cancer types reveals convergent and divergent mechanisms of bone colonization.* ***Cell Genomics*** https://doi.org/10.1016/j.xgen.2025.100888 (*equal contribution)
5. Xu, Z.*, **Liu, F.*,** Ding, Y., , Hao, X., Pan, T., Miles, G., Wu, Y.-H., Liu, J., Bado, I. L., Zhang, W., Wu, L., Gao, Y., Yu, L., Edwards, D. G., Chan, H. L., Aguirre, S., Dieffenbach, M. W., Chen, E., Shen, Y., Hoffman, D., Becerra-Dominguez, L., Rivas, C. H., Chen, X., Wang, H., Gugala, Z., Satcher, R. L., Zhang, X. H.-F. (2024) *Unbiased metastatic niche-labeling identifies estrogen receptor-positive macrophages as a barrier of T cell infiltration during bone colonization.* Preprint at: https://doi.org/10.1101/2024.05.07.593016 (*equal contribution) ***Cell*** Accepted
6. **Liu, F.**, Zhang, X. H.-F. (2025) *Decoding Metastatic Microenvironments Through Single-Cell Omics.* ***PLoS Biology*** 23(7): e3003299. https://doi.org/10.1371/journal.pbio.3003299

Other publications

7. Hao, X.*, Shen, Y.*, Chen, N., Zhang, W., Valverde, E., Wu, L., Chan, H. L., Xu, Z., Yu, L., Gao, Y., Bado, I., Michie, L. N., Rivas, C. H., Becerra Dominguez, L., Aguirre, S., Pingel, B. C., Wu, Y.-H., Liu, F., Ding, Y., Edwards, D. G., Liu, J., Alexander, A., Ueno, N. T., Hsueh, P.-R., Tu, C.-Y., Liu, L.-C., Chen, S.-H., Hung, M.-C., Lim, B., Zhang, X. H.-F. (2023) *Osteoprogenitor-GMP crosstalk underpins solid tumor-induced systemic immunosuppression and persists after tumor removal.* ***Cell Stem Cell***. 30(5):648-664.e8. http://doi.org/10.1016/j.stem.2023.04.005 PMID: 37146584; PMCID: PMC10165729. (*equal contribution)
8. Yuan, X., Hao, X., Chan, H. L., Zhao, N., Pedroza, D. A., **Liu, F.**, Le, K., Smith, A. J., Calderon, S. J., Lieu, N., Soth, M. J., Jones, P., Zhang, X. H.-F., Rosen, J. M. (2024) *CREB-binding protein/P300 bromodomain inhibition reduces neutrophil accumulation and activates antitumor immunity in triple-negative breast cancer*. ***JCI Insight***. 2024 Sep 17:e182621. http://doi.org/10.1172/jci.insight.182621 Epub ahead of print. PMID: 39287984.
9. Hao, X.*, Shen, Y.*, Liu, J., Alexander, A., Wu, L., Xu, Z., Yu, L., Gao, Y., **Liu, F.**, Chan, H. L., Li, C.-H., Ding, Y., Zhang, W., Edwards, D. G., Chen, N., Nasrazadani, A., Ueno, N. T., Lim, B., Zhang, X. H.-F. (2024) *Solid tumour-induced systemic immunosuppression involves dichotomous myeloid–B cell interactions*. ***Nature Cell Biology*** https://doi.org/10.1038/s41556-024-01508-6 (*equal contribution)
10. Pan, T., **Liu, F.**, Hao, X., Wang, S., Wasi, M., Song, J. H., Lewis, V. O., Lin, P. P., Moon, B., Bird, J. E., Panaretakis, T., Lin, S.-H., Wu, D., Farach-Carson, M. C., Wang, L., Zhang, N., An, Z., Zhang, X. H.-F., Satcher, R. L. (2024) *BIGH3 mediates apoptosis and gap junction failure in osteocytes during renal cell carcinoma bone metastasis progression*. ***Cancer Letters***. 596:217009. http://doi.org/10.1016/j.canlet.2024.217009 Epub 2024 Jun 5. PMID: 38849015.
11. Yu, L., Liebenberg, K., Shen, Y., **Liu, F.**, Xu, Z., Hao, X., Wu, L., Zhang, W., Chan, H. L., Wei, B., Lorenzi, P. L., Gao, Y., Bado, I., Becerra-Dominguez, L., Rivas, C. H., Aguirre, S., Pingel, B. C., Wu, Y. H., Ding, Y., Liu, J., Zhang, X. H. (2025) *Tumor-derived arachidonic acid reprograms neutrophils to promote immune suppression and therapy resistance in triple-negative breast cancer.* ***Immunity*** 58(4): 909–925.e7 https://doi.org/10.1016/j.immuni.2025.03.002
12. Han, Y., Sakar, H., Xu, Z., Lopez-Darwin, S., Wei, Y., Hang, X., **Liu, F.**, Tran, K., Wang, W., Miller, J. M., DeCoste, C. J., Blohm, D. S., Satcher, R. L., Zhang, X. H.-F., Kang, Y. (2025) *Tumors hijack macrophages for iron supply to promote bone metastasis and anemia* ***Cell*** https://doi.org/10.1016/j.cell.2025.08.013
13. Wu, L., Xu, Z., Wang, J., Ding, Y., Gao, Y., **Liu, F.**, Wang, S., Liu, J., Chan, H. L., Wu, W., Wu, Y.-H., Yu, L., Hao, X., Li, X., Edward, D. G., Sha, C., Lee, T. D., Aguirre, S., Becerra-Dominguez, L., Hoffman, D., Chen, X., Rivas, C. H., Bado, I., Zhang, W., Zhu, Q., Sreekumar, A., Satcher, R. L., Zhang, X. H.-F. (2025) *Symbiotic adhesion of neutrophils and epithelial-like cancer cells shapes the tumor microenvironment and promotes metastasis*. **Under Revision**
14. Pedroza, D., Yuan, X., **Liu, F.**, Chan, H., Zhang, C., Bowie, W., Smith, A., Calderon, S., Lieu, N., Wu, W., Porter, P., Sarkar, P., Zhao, N., Oehler, C., Peller, O., Gaber, W., Zhu, Q., Perou, C., Zhang, X., Rosen, J. (2025) *Anti-CSF-1R therapy with combined immuno-chemotherapy coordinate an adaptive immune*

response to eliminate macrophage-enriched triple-negative breast cancers. ***Nature Communications*** 10.1038/s41467-025-67964-2

15. Yu, L., Rivas, C. H., **Liu, F.**, Shen, Y., Xu, Z., Wu, L., Ding, Y., Hao, X., Zhang, W., Chan, H. L., Liu, J., Wei, B., Gao, Y., Becerra-Dominguez, L., Wu, Y.-H., Wang, S., Lee, T., Li, X., Edwards, D. G., Zhang, X. H.-F. (2025) *Tumor associated macrophages hijack the inflammation resolution pathway to escape therapy-induced immunosurveillance* ***JCI*** 10.1172/JCI198815

Poster and Conferences

2024	Cancer and Cell Program Symposium, BCM	Poster	2nd place
2025	Cancer and Bone Society Symposium	Presentation	

Reviewing activities

2024-present	Active Reviewer	*Nature Reviews Cancer* *Cancer Gene Therapy*

Teaching & Leadership

2023-2024 Monash University
Course: BMS5007 Biotechnology
Instructor: Dr. Yuzhou Wu
▶ Held online office hours and graded homework/exams.

2020 New York University
Course: CORE-UA-303 Human Genetics
Instructor: Dr. David Fitch
▶ Led weekly lab and recitation sessions for a class of 40 students.
▶ Held weekly Q&A sessions and graded assignments.

Data Analysis Skills

- Bulk RNA sequencing
- Single-Cell Sequencing
- Trajectory inference (including monocle3, RNA velocity)
- Single-cell pseudo-bulk transcriptomic comparisons
- Pathway enrichment analysis
- Genome analysis
- Special transcriptomic
- Multiplexed imaging analysis
- Computation language: R, Linux, and Python

Experimental Skills

- Molecular clone
- Mammalian cell culturing
- Flow cytometry analysis
- Protein expression, purification, and *in vitro* folding
- Mouse surgery (tumor implementation, tumor removal)
- Molecular analysis (PCR, qPCR, WB, BCA assay)
- Working with mouse models
- Confocal imaging

Activities

2015 Amazon Jungle Conservation Project
Chontachaka, Peru
▶ Identified and documented animal species; relocated endangered plants to designated preservation fields.